Assessment of SIMBox Fraud: An Approach to National Security Threat

Chapter · November 2021

DOI: 10.9734/bpi/ist/v1

1 author:

Kala Baskar
University of Madras

Print ISBN: 978-93-5547-131-4, eBook ISBN: 978-93-5547-209-0

Assessment of SIMBox Fraud: An Approach to National Security Threat

N. Kala [a#*]

DOI: 10.9734/bpi/ist/v1/12787D

ABSTRACT

SIMBox fraud is one of the most sophisticated fraud types in the recent times. Telecom regulators and mobile operators are facing massive revenue losses as bypass fraud continues to be one of the most prolific and costly frauds. To terminate international inbound calls to local subscribers, gateway equipment such as fixed, Voice over Internet Protocol (VoIP), Global System for Mobile Communication (GSM), Code Division Multiple Access (CDMA), VOIP to GSM, and fixed line gateway are used. Fraudsters bypass the connections by diverting traffic away from legal interconnect gateways. Fraudulent Operators sending outbound international traffic connect to interconnect operators with lower rates, leading to loss of revenue for the original internet service provider. SIMBox fraud is considered illegal since those who undertake it are not licensed to provide telecommunication services. SIMBox fraud is also considered as a national security threat since terrorist groups use this device to make international calls at local call rates. The main objectives of this study is to provide background of cellular networks, internet gateway, voice over internet protocol, preliminary overview to SIMBox architecture, internet bypass fraud - including GSM gateway fraud and SIMBox fraud and the State-of-Art Procedures to combat SIMBox fraud. This chapter focuses on the study of SIMBox fraud as a National Security threat. Further this study also suggests the methods for mitigating such security threat and the next generation forensic detection methods.

Keywords: Internet Bypass Fraud; SIMBox fraud; voice over internet protocol; call data analysis and management system; national security threat.

1. INTRODUCTION

1.1 Bypass Fraud [1]

A call via a legitimate path/route will be by passed, resulting in a revenue loss. Rates for making national or international calls are typically set by a country's regulators or by an individual or group of operators. Bypass fraud is common in countries where there is a difference in calling rates between national and international calling. Furthermore, government operators monopolise international gateways in some countries. The fraudsters take advantage of the difference in rates to ensure that they make enough profit, which serves as the primary motivator for them to invest in procuring the equipment and GSM connections needed to conduct a large-scale Bypass fraud. In countries where the international to national terminating charge margins are low, nil or negative, the bypass fraud either does not exists or is conducted on a very low scale. It is one of the latest and most severe threats to a telecom operator’s revenue. It is an unauthorized exploitation or manipulation of an operator’s network. This can happen in two ways:

- SIM Box Interconnect Fraud
- GSM Gateway Fraud

[#] *Assistant Professor*
[a] *Centre for Cyber Forensics and Information Security, University of Madras, Chennai, Tamilnadu, India.*
**Corresponding author: E-mail: kalabaskar@gmail.com;*

Through such methods fraudsters gain incentives to evade such high tariff interconnects and deliver costly international calls illicitly. Fraudsters use Voice over Protocol – Global System for Mobile Communications (VOIP-GSM) gateways, also called as “SIM Boxes”, which are used to receive incoming calls (via wired connections) and deliver them to a cellular voice network. It appears as if it is a local call appearing from a legitimate customer’s phone. This practice not only dramatically degrades the network experience for legitimate customers violating the telecommunication laws in many countries, but also is extremely profitable for SIMBoxers/fraudsters resulting in revenue loss significantly [2].

1.2 Cellular Networks

Commonly used standard for implementing cellular communication is a set of Global System for Mobile Communication (GSM). Majority of countries such as United States, Europe, Africa and Asia use GSM for mobile communication and is popularly called as 2G cellular networks and subsequently evolved into Universal Mobile Telecommunications Service (UMTS-3G) and Long Term Evolution (LTE-4G). A smart card called Subscriber Identity Module (SIM) is being used by GSM it carries the identity of subscriber (Integrated Circuit Card Identity - ICCID, International Mobile Subscribers Identity -IMSI) and is used for authenticating the user with operator. Every communication transaction in network is cryptographically authenticated. SIM cloning was prevalent in the recent past which negated guarantee of specific SIM card attribution. Latest advanced technology SIM cards have hardware security and practical key recovery protections that prevents card cloning. Additionally, GSM standards use audio codec called GSM Full- Rate (GSM-FR) which is also frequently implemented in Voice over Internet Protocol (VOIP) software [3].

1.3 Voice over Internet Protocol

Voice over Internet Protocol [4] (Voice over IP, VoIP and IP telephony) is a methodology and group of technologies for the delivery of voice communications and multimedia sessions over Internet Protocol (IP) networks. The terms Internet telephony, broadband telephony, and broadband phone service specifically refer to the provisioning of communication services (voice, fax, SMS, voice-messaging) over the public Internet, rather than via the Public Switched Telephone Network (PSTN). There are two ways in which clients can complete a VoIP call. Firstly, a call can be completed exclusively using internet. Secondly calls might also be routed from/to a VoIP client to a Public Switching Telephone Network (PSTN) through a VoIP gateway. The transport is similar to traditional telephony network. It uses special media codec’s protocol to encode audio and video. Popular VoIP providers are Skype, Vonage and Google which use both IP-only and IP-PSTN calls.

Step by step process for a VoIP call is given below:

i. VoIP calls are setup using text based protocol called as Session Initiation Protocol (SIP).
ii. SIP function is to establish which audio codec will be used for the call.
iii. Once a call connection has been established audio flows between the callers
iv. It uses Realtime Transport protocol (RTP) for this purpose
v. This is typically carried over User Datagram Protocol (UDP)
vi. Many codec’s are mandatory, although some are optional.
vii. GSM-FR is implemented outside of cellular network and is sometimes used by VoIP software.

2. SIMBox

SIMBox [5] shown in Fig. 1 is a device used as part of a VoIP gateway installation. It contains a number of SIM cards, which are linked to the gateway but housed and stored separately from it. A SIM box can have SIM cards of different mobile operators installed, permitting it to operate with several GSM gateways located in different places. The SIMBox operator can route international calls through the VoIP connection and connect the call as local traffic, allowing the box’s operator to bypass international rates and often undercut prices charged by local mobile network operators who connect VoIP calls to GSM voice network. It does not use data network. SIMBox device requires one or more

SIM cards to wirelessly connect VoIP call to GSM network. A SIMBox acts as a VoIP client whose audio input and output are connected to a Mobile Phone. These devices have a strong market in private enterprise telephone networks. Such private enterprise use GSM gateways with the permission of the licensed telecommunications service provider at lower cost for terminating a call. However, this is possible and legal only for domestic calls. It is enabled by Voice over Internet Protocol (VOIP) Global System for Mobile Communication (GSM). The most common implementation of interconnect bypass fraud is known as SIMBoxing. Enabled by VoIP GSM gateways (i.e., “SIMBoxes”), SIMBoxing connects incoming VoIP calls to local cellular voice network via a collection of SIM cards and cellular radios (9). In a normal course, the calls will be received by the network service provider and call tariffs will be charged. In SIMBoxing, calls will bypass the normal course of connection, appearing to originate from customer phone, to a network provider. The calls are delivered at a subsidized domestic rate instead of international rate. Such an activity has its negative impact on availability, reliability and quality of service for legitimate consumers. Moreover, it also creates network hotspots by injecting huge volume of tunneled calls, bandwidth restriction thereby causing revenue loss to network operators.

Fig. 1. SIMBox [6]

2.1 SIMBox Architecture

The Fig. 2 represents the SIMBox Architecture.

A. The GSM Gateway

This component allows call termination from the VoIP network to the cellular network and vice versa. It receives VoIP traffic from a softswitch, transcodes it, and ensures its routing on the wireless cellular network. At this end, it supports several VoIP signaling protocols, including SIP and H.323, and several voice codecs.

B. The SIMBank

The role of the SIMBank is to hold a bundle of SIM cards used by the system for call routing. The SIMBank is not essential at the SIMBox architecture because some GSM gateway models integrate SIM slots (as aforementioned). However, in contrast to GSM gateways, the SIMBank has the advantage of offering remote operation capability, which eases management tasks, minimizes the maintenance expenses, and solves the SIM-blocking issue. A SIMBank allows to install and manage SIM cards of different mobile operators and enhance the working of several GSM gateways placed in different locations. Depending on its model, a SIMBank can comprise between 32 to 256 SIM slots. Nevertheless, several SIMBanks can connect to one system providing the ability to use practically unlimited SIM cards.

C. The Control Server

The control server is the central component of the architecture. It is composed of a Linux-based core with web and database systems. It is often hosted on a Dedicated Server or on a Virtual Private

Server (VPS) on the cloud to be always available and easily accessible. Phonerlite, asterisk, goautodial are applications that mainly use control Servers for this purpose. In addition to this, a few other open-source software are also used as control servers. Communication diagram comprising of main stages of a call flow through SIMBox functional components is illustrated in Fig. 3.

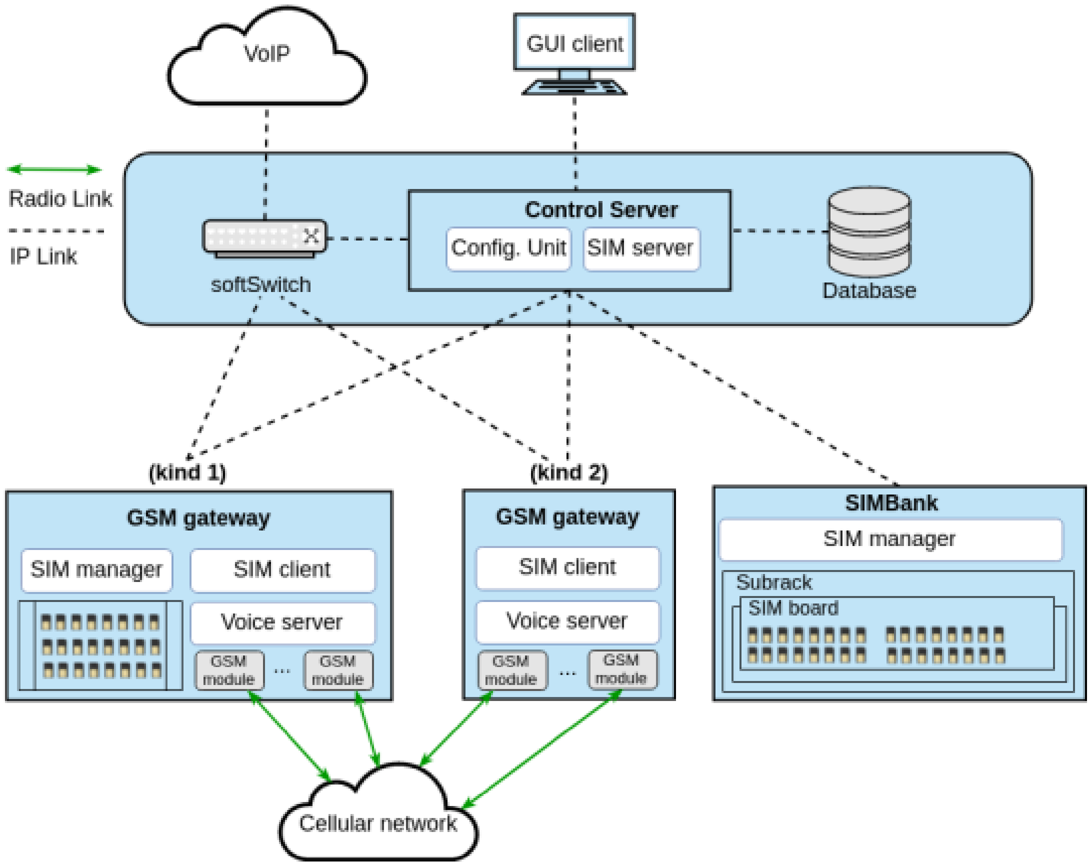

Fig. 2. Represents a fragmented architecture as all the different components of the SIMBox are distinguished [7]

D. The Control Server

The control server is the central component of the architecture. It is composed of a Linux-based core with web and database systems. It is often hosted on a Dedicated Server or on a Virtual Private Server (VPS) on the cloud to be always available and easily accessible. Phonerlite, asterisk, goautodial are applications that mainly use control Servers for this purpose. In addition to this, a few other open-source software are also used as control servers. Communication diagram comprising of main stages of a call flow through SIMBox functional components is illustrated in Fig. 3.

E. Interaction and Organization

The communication diagram of Fig. B summarizes a call termination flow within SIMBox components. The following steps can be identified in the Fig. 3.

As a prerequisite step, voice channels are continuously created within the system through SIM clients' requests to the control server

(1) A call comes from the external VoIP network to the softswitch.

(2) The softswitch sends the routing request for the call to the Control server.
(3) After processing the request, the control server responds with the appropriate route (voice channel), if any. If not, the call is dropped. The voice channel is chosen according to the policy defined in the control server's Role 3.
(4) The softswitch routes the call to the voice server of the selected voice channel.
(5) The voice server terminates the call to the cellular network.
(6) At last, the call connection is completed.

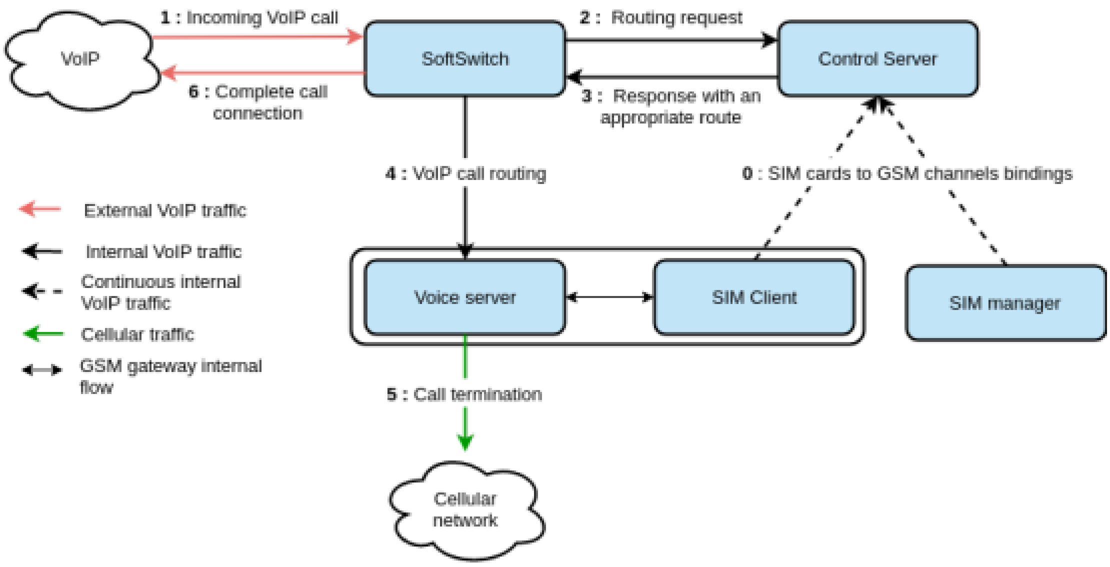

Fig. 3. Communication diagram: Main stages of a call flow through SIMBox functional components [8]

For establishing a parallel telephone exchange, fraudsters require a high-speed internet connection, basic networking devices, SIMbox with capability multiple SIMS (more than 500), Laptops or desktop computers, servers and controlling applications. Fraudsters usually rent a commercial place in Tier 2 cities such as Kozhikode in Kerala so that they can bypass call behavior analysis of Telecom Service providers and also evade lawful communication monitoring system.

2.2 Interconnect Bypass Fraud using SIMBoxing

Most common implementation of interconnect bypass fraud [9] is known as SIM Boxing. Fraudsters use SIMBox bypass the international calls and make it appear as if it is a domestic call causing revenue loss to telecom operators. There is a high demand for GSM-VoIP gateways spanning a wide range of features; numbers of concurrent calls are supported. Some of them have only limited functionality, while others hold several SIMcards and also supports a variety of audio codecs in a "SIM server". Sometimes one or more radio interfaces calls using the "Virtual SIM cards" from the server. This prevents location based fraud detection. Miscreants, utilize this and commits the fraud. The cost of SIMBox equipment goes upto 200,000 USD. A typical international call which is routed through a regulated licensed interconnect is illustrated in the Fig. 4. Let us assume client 'A' is located in India and client 'B' is located in UK. In a typical call, when client 'A' is calling client 'B', the call is routed through the telephone network in India (labeled as "Foreign PSTN core") to an interconnect between client 'A' and client 'B' network in UK. This passes through client B's domestic network (labeled as "Domestic PSTN Core") and communication establishes between client 'A' and client 'B'. If client 'A' and client 'B' are not in neighboring countries, there can be many interconnects and intermediary networks. This is very critical the connections are heavily monitored for billing purpose and quality. It can be seen that VoIP calls initiated from services such as Skype that terminates on a mobile phone also passes through regulated interconnect. A SIMBoxes international call avoids regulated interconnect by routing the call to a SIMBox which completes the call using the local cellular network. In a SIMBox case, client ‘A’ call is routed through domestic network, but instead of passing through the regulated interconnect, the call is routed over internet protocol (VoIP) to SIMBox in the destination country. In doing so, the SIMBox places a separate call on the cellular network in the destination country, then routes the audio from IP call into the cellular call, which is routed to client ‘B’ through the domestic network. The same is illustrated in Fig. 5. The main disadvantage here is neither of end users is aware that the call is being routed through a SIMBox. This causes a contractual breach of trust between two Internet Service Providers (ISPs) who have agreed to route traffic between their networks. The intermediaries own profit from reduced prices. Two types of attack can take place. Firstly, hijacking of international call; secondly, hijacking and re-injecting of an international call. First type has been described above. In the second type, SIMBoxes re-inject telecom voice traffic into the mobile network masked as mobile customers and operator has to pay for the re-injected calls.

Ingeneral there are three types of routes that are used in communicationnetworks. They are:

i. **White Route**: both source and destination have legaltermination.
ii. **Black Route**: both source and destination have illegaltermination.
iii. **Grey Route**: the termination is legal for one entity or country,but illegal for the other end.

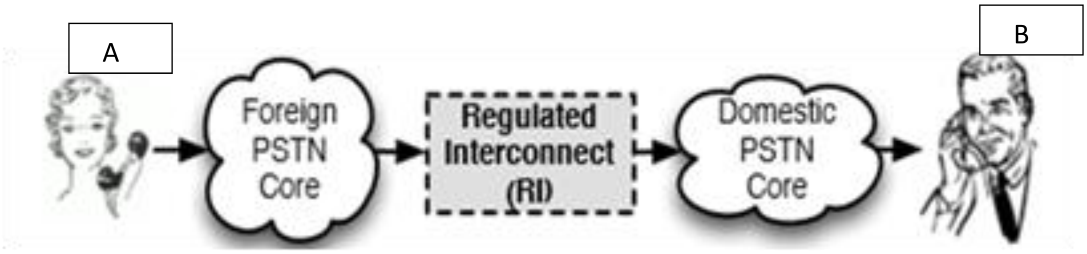

Fig. 4. Typical international call routed through regulated licensed interconnect [7]

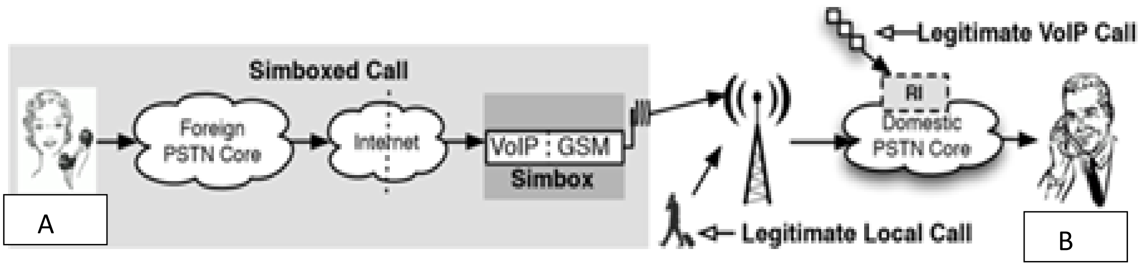

Fig. 5. A SIMBox International Call [7]

2.3 GSM Gateway Interconnect Devices

Interconnect systems, such as gateways, allows voice interoperability between otherwise incompatible radio communications systems. Interoperability is achieved by retransmitting voice over interconnected radio subscriber both mobile as well as portable units. Linking incompatible radio frequency bands and systems can be relatively easy and effective. Interconnect deployment requires a new strategy and operational procedures. The gateway approach to interoperability has significant potential, considering the ease of gateway deployment and relatively low cost when compared to wide area radio system. A gateway is a type of interconnect system. They can also connect trunked talk groups, encrypted networks, public telephone systems, and cellular or satellite phone connections. Most gateway devices are mobile and portable, but many are used in permanent configurations [10].

2.4 Interconnect Bypass Fraud Global Scenario

According to a survey conducted by Communication Fraud Control Association (CFCA), in the year 2015, the revenue loss amounts to about $3.77 Billion USD. According to this survey, top 10 countries where the fraudulent calls originated, is listed in Table 1. Further survey points out the percentage of top five frauds, in which interconnect Bypass fraud in network is around 5%, whereas in roaming status, interconnect bypass fraud amounts between 20 – 25%. [11]. This can be seen evidently from the following Fig. 7. Authorities in US say that hackers were involved in an international crime ring that scammed telecommunication companies out of an estimated $50million USD in last few years. FBI most wanted list of cyber criminals have been arrested by authorities in their native Pakistan. Serbian Police cracks down on illegal SIM Box Scheme. According to Serbia's interior Ministry in cooperation with the special department of cybercrime of Prosecutor's Office and the Ministry of Interior Macedonia have identified miscreants using SIMBoxes to bypass international communications via VoIP and making low-cost calls in Serbia. More than 40,000 SIM cards were found in Macedonia of mobile operators from Serbia, Croatia, Slovenia, Albania, Bosnia and Herzegovina. There are incidents in Ghana [12] where the fraudsters connived with partners abroad to route internet calls via VoIP to make it appear as if the call is a local one. Even. The seized SIMcards and connecting devices are illustrated in Fig. 6. There has been incidence where even women have been arrested for alleged SIMBox fraud.

2.5 Interconnect Bypass – SIMBox Fraud Indian Scenario

Table 2 illustrates different SIMBox cases, reported state (state, date and month), the devices that were seized, investigating agencies and the impact of such crime.

Table 1. Country wise fraudulent calls in percentage based on call origin.

Countries	Fraudulent calls percentage
United states	5%
Pakistan	4%
Spain	4%
Cuba	3%
Italy	3%
Philippines	3%
Somalia	3%
United Kingdom	2%
Dominican Republic	1%
Egypt	1%

Fig. 6. Seized SIM cards and SIMbox [13]

Table 2. Interconnect bypass fraud using SIMboxes in India

S. No	Source of information	Details of the case	Reported State	Reported Month and Year	Devices seized	Investigaio n Agency	Impact of fraud
1	Times of India[14]	Anti-Terrorism squad have busted an illegal telephone exchange and spying racket causing national security threat. This act has been committed by a software engineer from south Delhi and ten others from Lucknow and other parts of UP. The exchanges were not only making lakhs of rupees by routing international calls bypassing the legal gateways. These systems were used for Pakistan's Inter- Service Intelligence (ISI) to call Army officials to elicit information from them.	NCR	Jan-17	16 SIM BOX units, 140 prepaid cards, 10 mobile phone and 28 data cards and five laptops	ATS Delhi and UP Police	Revenue loss and National Security
2	The News Minutes [15]	The Crime Branch conducted raids six locations in Bengaluru following intelligence inputs by Military Intelligence (MI) on illegal calls made to the Indian Army helpline numbers in West Bengal. The bust uncovered an illegal telephone network to route calls from operatives of terror outfits. The joint operation by the MI Southern Command and Bengaluru Police Crime Branch (ATC) led to the arrests of two suspects	Karnataka	Jun-21	3,000 SIM cards, 109 SIM box devices, 23 laptops, 10 pen drives, 14 UPS and 17 routers	Bengaluru Crime branch and ATC	Revenue loss and National Security
3	New Indian Express [16]	CCB sleuths have unearthed a SIM box fraud and arrested a Kerala-based man who was running the illegal telephone exchange at a rented house in Mysore.	Karnataka	Jul-21	4 SIM boxes, 500 Sims, several cellphones, computers, Wi-Fi routers with SIM slots and modems.	Mysore City Police	Revenue loss

4	Times of India [17]	Based on inputs from a J&K based Military Intelligence unit, the Telangana Police have busted eight illegal VoIP (voice over internet telephony) exchanges in Hyderabad that were reportedly being used by Pakistan intelligence agencies for espionage operations in India	Telangana	Apr-17	30 SIM boxes and more than 650 SIM cards	Telangana Police	Revenue loss and National Security
5	The Hindu [18]	The Crime Branch conducted raids following intelligence inputs by Jammu & Kashmir's Military Intelligence (MI) on illegal calls being made to the Army's civil exchange seeking sensitive information like deployment.	Maharashtra	May-20	five SIM boxes, of which four were operational, and 223 SIM	Mumbai Police	Revenue loss and National Security
6	Times of India[19]	Police constituted a special investigation team (SIT) after a network of illegal telephone exchanges that routed international calls by converting them to local calls was busted in Kozhikode.	Kerala	Jul-21	730 SIM cards, 26 SIM boxes and 24 routers	SIT	Revenue Loss
7	Ahmedabad Mirror [20]	The Kolkata Special Task Force arrested three persons, including one Bangladeshi National identified as Mamun, from the airport area	West Bengal	Sep-21	23 SIM boxes, 400 pre-activated SIM cards, Wi-Fi modems of different service providers, laptops and other communication equipment	Kolkata Special Task Force	Revenue loss and National Security

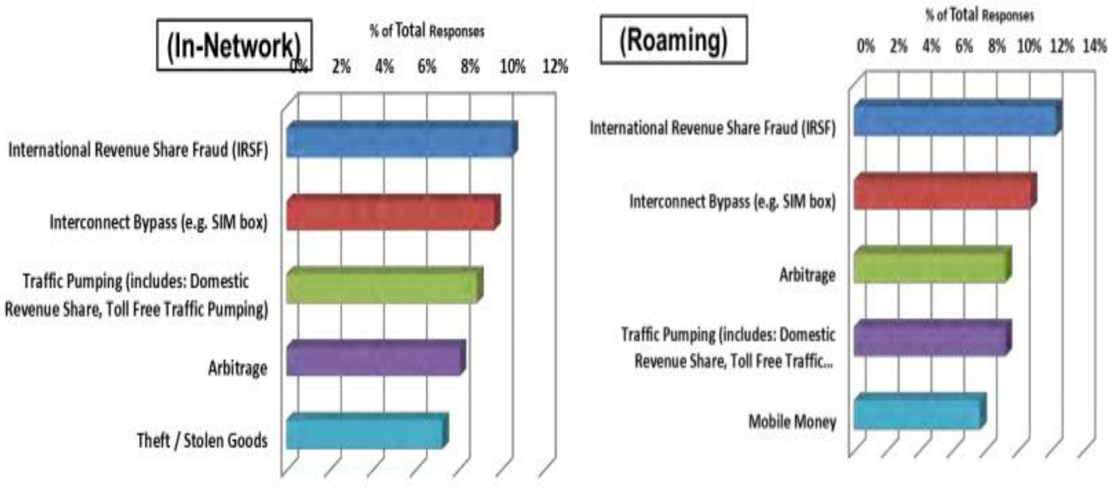

Fig. 7. Percentage comparison of SIMBox fraud in network vs. roaming [13]

2.6 Impact of SIMBox Frauds

A. Financial & Revenue Loss

• Tax Revenue Loss — Bypass fraud causes huge loss in the international voice call taxes used to fund national infrastructure and other government programs.

• Operator Revenue Loss — Unlicensed SIM Box operators rob the licensed telecom operators of their livelihood, jeopardizing their ability to compete and serve customers well.

B. National Infrastructure & Service Quality Damage

• Damage to National Infrastructure – When bypass fraudsters destroy the incentive for licensed operator to invest in their networks, the whole nation feels the impact of degraded communication infrastructure.

• Quality of Service Declines — Fraudsters lower the average QoS of a nation's voice service because they use cheap, low-quality equipment to cut their costs.

C. Security & Privacy Damage

• Lawful intercept systems are bypassed — Illegal SIM Box terminated calls sidestep the lawful surveillance systems that police, and intelligence agencies use to track criminals and terrorists.

• Voice Privacy and Security Protections -- Public networks have security, privacy, and encryption built in, but bypass networks are often sent in the clear allowing criminals or hackers to listen in on conversations.

• SMS messages are compromised — SMS messages are also compromised by bypass. This is especially troubling since SMS has become a major communication channel for confidential and personal data. Application-to-Person (A2P) SMS termination is a huge growth area fueled by automated notification systems such as those within the banking, airline, and consumer retail markets.

2.7 Countermeasures for Interconnect Bypass Fraud

Most common approaches to detect Bypass fraud are:

i. Bypass route detection through call generation
ii. SIMBox detection using Fraud Management systems (FMS)
iii. Unusual flows and volumes
iv. Unusual called number spreads
v. A-typical traffic peaks for on-net traffic
vi. Many SIM card identities (IMSIs) to a single equipment identity (IMEI)
vii. Use of only one cell site
viii. An absence of SMS, data or roaming service use

FMS system includes the following detection systems:

i. Traffic analysis
ii. International Mobile Subscriber Identity (IMSI) number/ Integrated Circuit Card ID (ICCID) series analysis
iii. International Mobile Equipment Identity number (IMEI) [21]

2.8 Next Generation Forensic Detection Methods

2.8.1 Call Data analysis and management system (CDAMS)

CDAMS is a powerful CDR-Analysis tool with proven advanced capabilities for intelligent analysis and automated processing of huge volumes of CDR (Call Detail Record / Telephone Toll Record / Cell phone Tower Dump) data within an intuitive, user-friendly interface. CDAMS has a sophisticated multi-layered search query engine used for building comprehensive suspect profiles. CDAMS enables analysts and intelligence professionals to convert large volumes of complex, disparate data into actionable intelligence which includes integration of call detaild records, tower dumps, sim card extraction details imsi and TMSI Location, gateway data details, Geo fensing Subscribers details, recover data from forensic tools such as UFED, XRY and Tarantula.

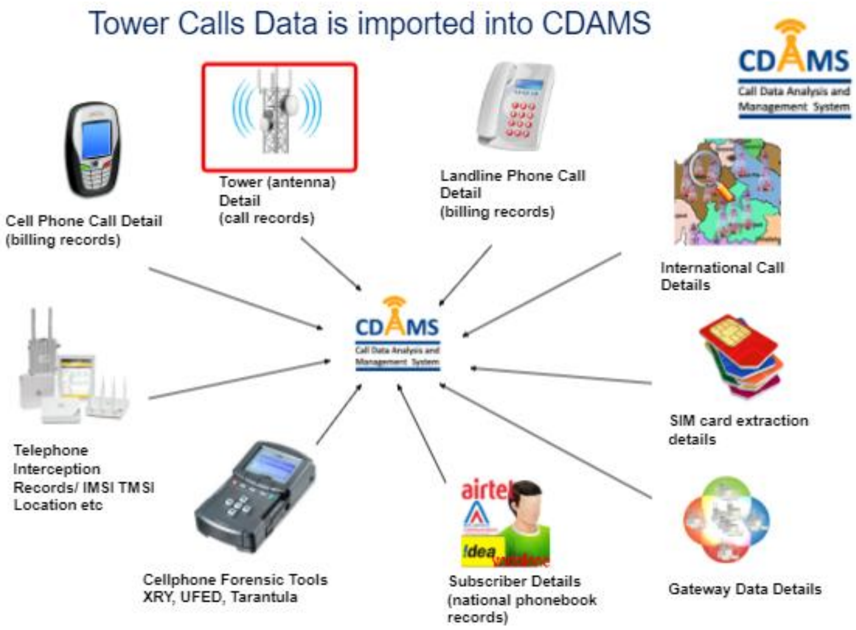

Fig. 8. Call Data analysis and management system (CDAMS) [22]

3. CONCLUSION

Telecommunication networks in developing nations rely upon the tariffs collected through regulated licensed interconnects in order to subsidize the cost of their deployment and operation. Fraudsters bypass the legal connection and commit fraudulent activity using SIMBoxes by tunneling traffic from VoIP connection into a genuine network unauthorized way. In this study an attempt has been made to identify the different ways interconnect bypass occurs. Further, this study has also identified the different devices used to commit such fraudulent activity. Moreover, a comparison between the current scenario from global perspective and Indian perspective has been made. Focus has been made to identify how the antisocial element and miscreants use such methodology and cause a national security threat across the globe. Finally, Fraud Management Systems that are currently in vogue and next generation forensic detection methods have been discussed.

COMPETING INTERESTS

Author has declared that no competing interests exist.

REFERENCES

1. Subex Limited. Bypass Fraud- Are you getting it right? [E-book]. Subex Limited; 2011. https://www.subex.com/pdf/Bypass_Fraud.pdf
2. Subex Limited. Telecom Fraud Alerts | Digital Fraud Management | Fraud News; 2020. https://www.subex.com/fraud-alerts/
3. Types of Telecom Frauds - Squire Technologies | Telecoms Signalling Specialists. (n.d.). Squire Technologies; 2021.
   Retrieved October 5, 2021
   https://www.squire-technologies.co.uk/types-of-telecom-frauds
4. Voice over IP. In Wikipedia; 2021.
   https://en.wikipedia.org/wiki/Voice_over_IP
5. SIM box. In Wikipedia. ; 2020.
   https://en.wikipedia.org/wiki/SIM_box
6. GSM SIM Bank 128 / 256 Sim Slots. (n.d.). Alibaba. Retrieved October 5, 2021,
   https://www.alibaba.com/product-detail/GSM-sim-bank-128-256-sim_60196031409.html
7. Kouam, AJ, Viana, AC, Tchana A. SIMBox bypass frauds in cellular networks: Strategies, evolution, detection, and future directions. IEEE Communications Surveys & Tutorials. Published; 2021.
   https://doi.org/10.1109/comst.2021.3100916
8. Kouam AJ, Viana, AC, Tchana A. SIMBox bypass frauds in cellular networks: Strategies, evolution, detection, and future directions. IEEE Communications Surveys & Tutorials. Published; 2021.
   https://doi.org/10.1109/comst.2021.3100916
9. Reaves B, Shernan E, Bates A, Carter H, Traynor P. Boxed out: Blocking cellular interconnect bypass fraud at the network edge. In 24th {USENIX} Security Symposium ({USENIX} Security 15) 2015;833-848.
10. US. Department of Justice. Interoperability Gateways/Interconnects (NCJ 217105 ed.) [E-book]. National Institute of Justice; 2007.
    https://www.ojp.gov/pdffiles1/nij/217105.pdf
11. Communications Fraud Control Association. (2019). Fraud Loss Survey.
    https://cfca.org/wp-content/uploads/2021/02/CFCA-2019-Fraud-Loss-Survey.pdf
12. U. $900,000 Goes To The Drain In Ghana Through SIM Box Fraud. Kofycity; 2016.
    https://kofycity.blogspot.com/2016/03/900000-goes-to-drain-in-ghana-through.html
13. U. (2016, March 30). $900,000 Goes To The Drain In Ghana Through SIM Box Fraud. Kofycity.
    https://kofycity.blogspot.com/2016/03/900000-goes-to-drain-in-ghana-through.html
14. Shalabh. Uttar Pradesh ATS busts international call racket spying on Army units, 11 arrested. Times of India; 2017.
    https://timesofindia.indiatimes.com/india/uttar-pradesh-ats-busts-international-call-racket-spying-on-army-units-11-arrested/articleshow/56781030.cms

15. IANS. Bengaluru police arrest five more in illegal SIM box exchange case. The News Minute; 2021. https://www.thenewsminute.com/article/bengaluru-police-arrest-five-more-illegal-sim-box-exchange-case-150669
16. Karthik KK. Cops unearth SIM box fraud in Mysuru, Kerala man held. The New Indian Express; 2021. https://www.newindianexpress.com/states/karnataka/2021/jul/31/cops-unearth-sim-box-fraud-in-mysuru-kerala-man-held-2338019.html
17. Mengle SG. SIM box racket busted in Govandi, one arrested. The Hindu; 2020. https://www.thehindu.com/news/cities/mumbai/sim-box-racket-busted-in-govandi-one-arrested/article31713192.ece
18. Shekhar R. SIMbox fraud unearthed in Hyderabad, 4 arrested. Times of India; 2017. https://timesofindia.indiatimes.com/india/sim-box-fraud-unearthed-in-hyderabad-4-arrested/articleshow/58170773.cms
19. TNN. (2021, July 3). SIT to probe illegal phone exchanges busted in Kozhikode. Times of India. https://timesofindia.indiatimes.com/city/kozhikode/sit-to-probe-illegal-phone-exchanges-busted-in-koz/articleshow/84077059.cms
20. Agencies. Kolkata SIT Seizes 23 SIM Boxes, State Under Alert. Ahmedabad Mirror; 2021. https://ahmedabadmirror.com/kolkata-sit-seizes-23-sim-boxes-state-under-alert/81807737.html
21. Khan AM, Yasir Imtiaz S, Shakir M. (Eds.). Automatic Monitoring & Detection System (AMDS) for Grey Traffic (Vol. 2, Issue WCECS 2015). Proceedings of the World Congress on Engineering and Computer Science; 2015.
22. CDAMS Call Data Analysis and Management System. CDR. (n.d.). Retrieved October 5, 2021, https://www.cdr-analysis.com

Biography of author(s)

Dr. N. Kala
Centre for Cyber Forensics and Information Security, University of Madras, Chennai, Tamilnadu, India.

She has a Masters in Forensic Science and PhD in Cyber Forensics, formerly served as Senior Scientific Officer at the Forensic Sciences Department, Government of Tamilnadu for 23 years. She was appointed as Assistant Professor at the Centre for Cyber Forensics and Information Security, University of Madras in the year 2014 and served as Director at the Centre. She has brought out a book titled "Cyber Tales" for children dealing with Cyber Security Awareness. She organized a landmark program first of its kind in the country Cyber Security Awareness Program (CSAP) in several Schools and Colleges in which several students participated in this program. She has also conducted a Cyber Security Awareness Rally by giving awareness to children, college students, and also general public. She has a couple of National Award in the field of Cyber Forensics from Ministry of Home Affairs, India. She has trained several Judicial Officers, Law Enforcement Officers, Faculties, Officers from Auditor General Office, Air force Authorities in the field of Cybercrimes and Cyber Forensics. She has been an expert witness in many courts in the country. She has published many articles in journals in India and abroad. She has been invited as a distinguished speaker in the field of Cyber Security and Cyber Forensics in India and countries such as Malaysia. She is holding membership in many organization such as Data Security Council of India (DSCI), International Federation for Information Processing (IFIP), International Institute for Certified Forensic Professional (IICFIP), The Society of Digital Information and Wireless Communications (SDIWC) and Indian Society of Criminology. She has been a reviewer for many journals and a book. She has also visited countries like Austria, Germany, UK and US.

DISCLAIMER
This chapter is an extended version of the article published by the same author(s) in the following journal.
Forensic Research & Criminology International Journal, 7(1), 2019.